

CSCI3240 Introduction to Computer Vision

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Lab 01 - Image basics (numpy)

Goal: get comfortable treating images as numpy arrays.

Task 1. Load an RGB image with imageio (or PIL) into a numpy array; print shape, dtype, min and max.

Task 2. Convert the image to grayscale with the weights $0.299 R + 0.587 G + 0.114 B$ using numpy only.

Task 3. Crop the central 50% of the image and flip it horizontally with array slicing (no loops).

Task 4. Compute a 256-bin intensity histogram with numpy.bincount and plot it; comment on the contrast.

Task 5. Threshold the grayscale image at its mean intensity and count foreground pixels; save the mask as PNG.

Deliverable: a notebook lab01.ipynb with the five tasks, submitted on WSP before Friday 23:59 (week 2).